Career Launcher MBA

QA Algebra-07

Co-ordinate Geometry

Question 1

Two points A(3, 7) and B(–5, 13) are located on a coordinate plane. Find

(1) The distance between A, B

(2) The mid-point of AB

(3) Coordinates of C such that AC : CB is 1 : 2

(4) Equation of the line passing through A, B

Question 2

Find the condition under which the straight lines $x + 2y - 9 = 0$, $3x + 5y - 5 = 0$ and $ax + by - 1 = 0$ are concurrent.

(1) $35a + 22b = 1$

(2) $35a - 22b = -1$

(3) $35a + 22b = 0$

(4) $22a + 35b = -1$

Question 3

What is the perpendicular distance between the two parallel lines $3x + 4y + 3 = 0$ and $3x + 4y + 12 = 0$?

(1) $\frac{11}{5}$ units

(2) 2 units

(3) 3 units

(4) $\frac{9}{5}$ units

Question 4

The four distinct points (–a, –b), (0, 0), (a,b) and (a^2, ab) are

(1) vertices of parallelogram

(2) vertices of rectangle

(3) collinear

(4) lying on a circle

Question 5

Find the minimum possible distance between a point on the circle $(x - 1)^2 + y^2 = 9$ and the line $8x + 6y - 48 = 0$.

(1) 0 unit

(2) 1 unit

(3) 2 units

(4) 3 units

Question 6

Find the area enclosed by the graphs

(1) $|x| = 4$; $|y| = 3$

(2) $|x - 3| = 4$; $|y + 3| = 5$

(3) $|x + y| = 3$; $|x - y| = 5$

Question 7

Find the area of the region bounded by the curves $|x| = 3$, $y = |x|$ and $y = -|x|$.

(1) 8 square units

(2) 9 square units

(3) 12 square units

(4) 18 square units

Question 8

In the X-Y plane, find the area of the region bounded by the graph $|x + 2y| + |x - 2y| = 6$

(1) 18 sq. unit

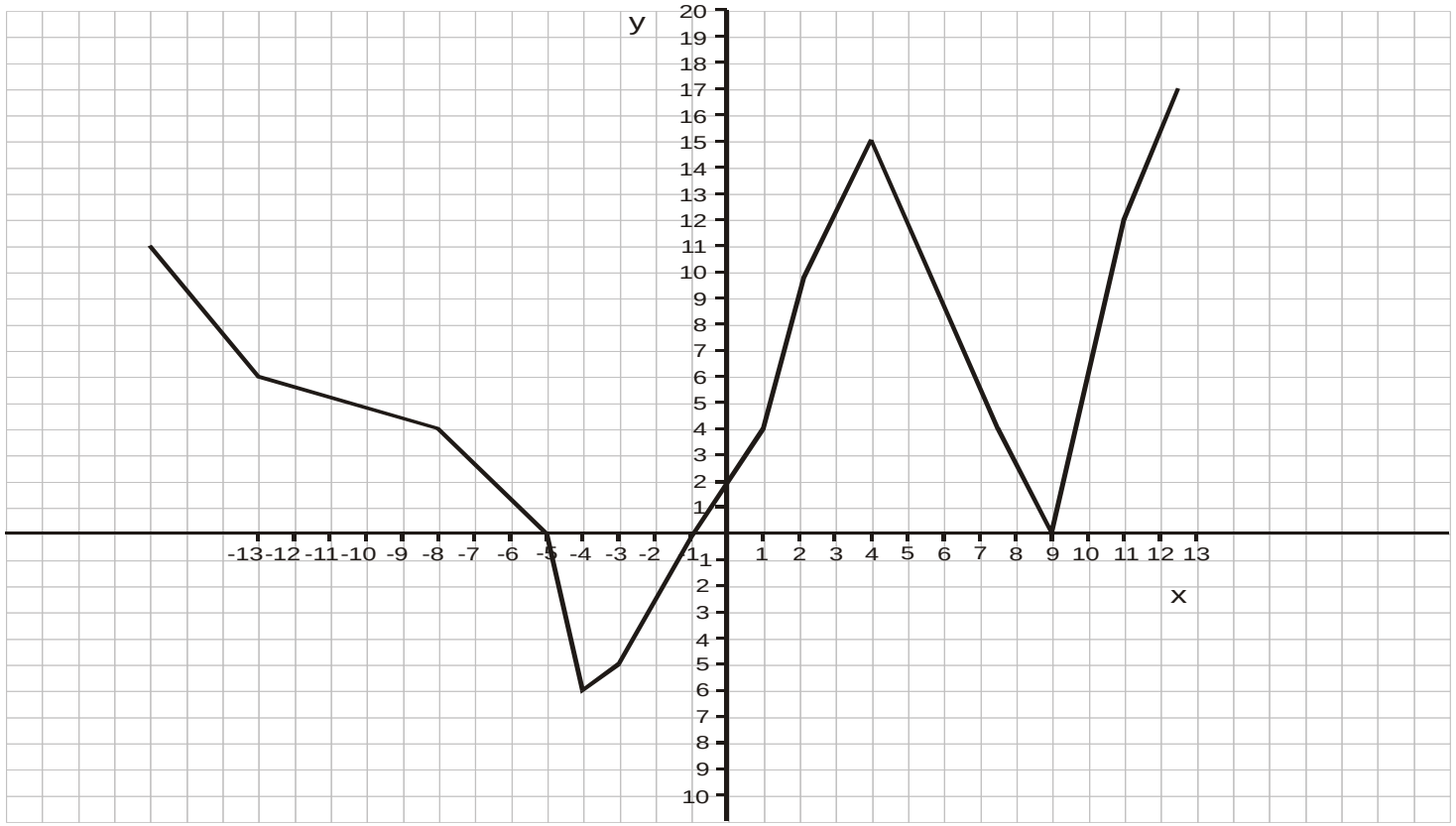
(2) 16 sq. unit

(3) 12 sq. unit

(4) 20 sq. unit

Question 9

The figure below shows the graph of a function f(x). How many solutions does the equation f(f(x)) = 15 have?



(1) 5

(2) 6

(3) 7

(4) 8

Question 10

In a rectangular coordinate system, points L, M, N and O are represented by the coordinates (–5, 0), (1, –1), (0, 5), and (–1, 5) respectively. Consider a variable point P in the same plane. The minimum value of PL + PM + PN + PO is

(1) $1+\sqrt{37}$

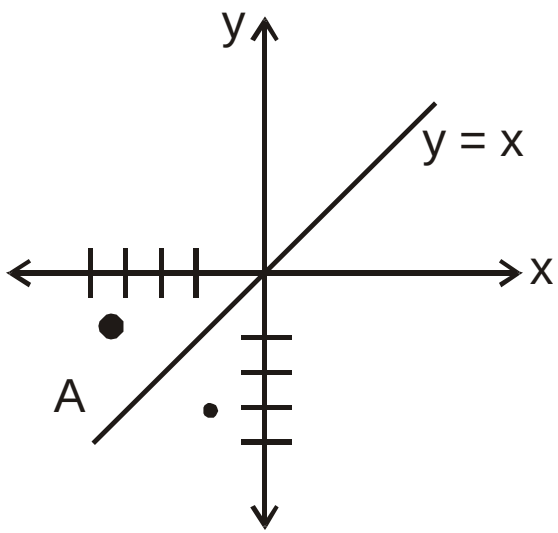
(2) $5\sqrt{2}+2\sqrt{10}$

(3) $\sqrt{41}+\sqrt{37}$

(4) $\sqrt{41}+1$

Question 11

In the figure given below, line y = x is the perpendicular bisector of line segment AB (not shown in the figure) and the x-axis is the perpendicular bisector of the line segment BC (not shown in the figure). If the coordinates of point A are (–4, –3), what are the coordinates of point C?



(1) (–3, – 4)

(2) (– 4, 3)

(3) (4, 3)

(4) (–3, 4)

Question 12

A line joining (5, 2) and (6, –15) subtends a right angle at which point?

(1) (5, 6)

(2) (2, 5)

(3) (2, 6)

(4) Insufficient Data

Question 13

Find the intercepts made by the line $3x + 4y - 12 = 0$ on the axes:

(1) 2 and 3

(2) 4 and 3

(3) 3 and 5

(4) None of these

Question 14

ABCD is a rectangle in the clockwise direction. The coordinates of A are (1, 3) and the coordinates of C are (5, 1), the coordinates of vertices B and D satisfy the line $y = 2x + c$, then what will be the coordinates of the mid-point of BC.

(1) (5/2, 7/2)

(2) (9/2, 5/2)

(3) (9/5, 7/2)

(4) (3, 2)

Question 15

ABCD is a parallelogram whose diagonals are parallel to the lines $2y - x - 5 = 0$ and $y + 2x - 7 = 0$ respectively.

Then ABCD is-

(1) Cyclic quadrilateral

(2) Rectangle

(3) Cube

(4) Rhombus

Question 16

If the co-ordinates of orthocentre and the centroid of a triangle ABC are (–5, 7) and (5, 5), then the circumcentre of the triangle ABC is:

(1) (25, 1)

(2) (10, 4)

(3) (–5, 2)

(4) (0, 6)

Question 17

The coordinates of a triangle ABC are A(1, 5), B(–2, 3), and C(0, –4); find the equation of the median AD?

(1) $7x - 3y + 8 = 0$

(2) $5x - 4y + 15 = 0$

(3) $x + 3y - 16 = 0$

(4) $11x - 4y + 9 = 0$

Question 18

The area of the closed region bounded by the equation $|x| + |y| = 2$ in the two-dimensional plane is

(1) 4π

(2) 4

(3) 8

(4) 2π

Question 19

The shortest distance of the point $\left(\frac{1}{2},1\right)$ from the curve $y = |x - 1| + |x + 1|$ is

(1) 1

(2) 0

(3) $\sqrt{2}$

(4) $\sqrt{\frac{3}{2}}$

Question 20

The points (2, 5) and (6, 3) are two end points of a diagonal of a rectangle. If the other diagonal has the equation $y = 3x + c$, then c is

(1) –5

(2) –6

(3) –7

(4) –8

Question 21

Let T be the triangle formed by the straight line $3x + 5y - 45 = 0$ and the coordinate axes. Let the circumcircle of T have radius of length L, measured in the same unit as the coordinate axes. Then, the integer closest to L is

Question 22

The vertices of a triangle are (0, 0), (4, 0) and (3, 9). The area of the circle passing through these three points is

(1) $\frac{205\pi}{9}$

(2) $\frac{123\pi}{7}$

(3) $\frac{14\pi}{3}$

(4) $\frac{12\pi}{5}$

Question 23

The points (2, 1) and (–3, –4) are opposite vertices of a parallelogram. If the other two vertices lie on the line $x + 9y + c = 0$, then c is

(1) 15

(2) 12

(3) 13

(4) 14

Question 24

Let ABCD be a parallelogram such that the coordinates of its three vertices A, B, C are (1, 1), (3, 4) and (–2, 8), respectively. Then, the coordinates of the vertex D are

(1) (–4, 5)

(2) (4, 5)

(3) (0, 11)

(4) (–3, 4)

Question 25

The area, in sq. units, enclosed by the lines $x = 2$, $y = |x – 2| + 4$, the X-axis and the Y-axis is equal to

(1) 6

(2) 8

(3) 12

(4) 10

Thanks!